

LABORATORY COURSE
CONTROL OF ELECTRIC DRIVES AND MACHINES
COURSE-Nr. 431.312

Unit 4:

Control of induction-machine

Teaching goals

Compare real world measurements with simulation results. Check if expected behaviour, e.g. design criteria, meet the measurable ones. Show that the more detailed model, compared to the transfer functions used for controller design, is able to give a proof for the proper functioning of the controllers.

In the lab

Apply current-, flux-, speed- and load reference steps and measure the system responses. Maximum stator current space vector 30A, maximum stator voltage space vector 55V.

1 Structure

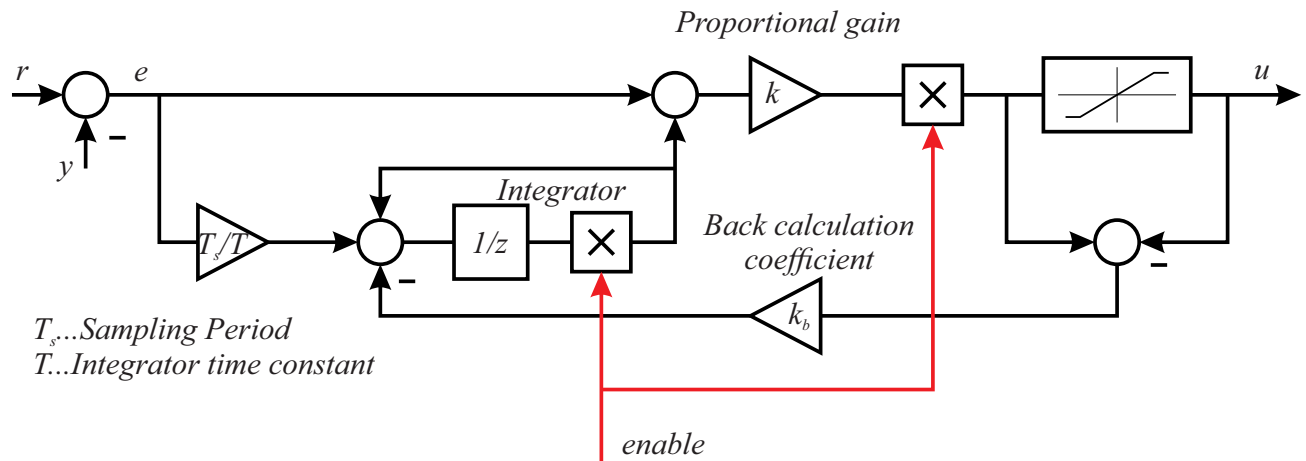


Figure 1: PI-Contoller

The PI-controllers z-transfer function is given by

$$C(z) = \frac{b_1 z + b_0}{z - 1} = b_1 \left[1 + \frac{(b_1 + b_0)}{b_1} \cdot \frac{1}{z - 1} \right] \quad (1)$$

Figure 1 shows the implemented controller structure. According to eqn. 1

$$k = b_1$$

$$\frac{T_s}{T} = \frac{b_1 + b_0}{b_1}$$

hold.

As long as the converter is turned off, the integrator and the output is kept at zero to avoid integration of measurement offset.

Please deliver controller coefficients as struct *ctrl* for the current controller with fields

ctrl.i.k...proportional gain
ctrl.i.T...integrator time constant
ctrl.i.kb...back calculation coefficient

with fields

ctrl.lambdar.k...proportional gain
ctrl.lambdar.T...integrator time constant
ctrl.lambdar.kb...back calculation coefficient

for the rotor flux controller and with fields

ctrl.w.k...proportional gain
ctrl.w.T...integrator time constant
ctrl.w.kb...back calculation coefficient

for the speed controller.

If your back calculation scheme does not coincide with the one given in fig. 1, attach a sketch of it, to enable correct interpretation of your back calculation coefficients.

2 Experiments

Inspection of step responses of the controlled induction-machine.

Nr.	DC-machine	Induction machine
	n_{ref} rpm	reference A or Vs
1	0	i_{sd} 0 $\rightarrow$ 10 A
2	0	λ_{rd} 0 $\rightarrow$ 0.08 Vs
3	0	i_{sq} 0 $\rightarrow$ 10 A
4	0	i_{sq} 10 $\rightarrow$ 0 A
5	1000	i_{sq} 0 $\rightarrow$ 10 A
6	1000	i_{sq} 10 $\rightarrow$ 0 A
	i_{aref} A	n_{ref} rpm
7	0 or off	0 $\rightarrow$ 100*
8	0 or off	0 $\rightarrow$ 1000
9	0 $\rightarrow$ -15	200
10	brake	-2000 $\rightarrow$ 2000

Table 1: Experiments; $x \rightarrow y$ stands for a setpoint step change from value x to value y ; off means turned off converter; brake stands for the usage of the converter as a brake chopper. 100* means a reference speed so that current limitation is not activated.

1. Only d-current controller is enabled.
2. Flux- and d-current controllers are enabled. Due to asymmetry q-current deviates from zero. This experiment could also be done with enabled q-current controller.
3. Q-current-, flux- and d-current controllers are enabled. Torque is developed, stator current space vector rotates.
4. When q-current and speed arrive at zero, stator space vector is at standstill.
5. If system is linear, it should not matter if speed equals zero or 1000 rpm.
6. If system is linear, it should not matter if speed equals zero or 1000 rpm.
7. Linear behaviour is expected, step response should look like it was designed for. Adjust the speed step according to maximum i_{sq} reference current $n_{ref} < 30/\pi \cdot \sqrt{(i_{max}^2 - i_{sd}^2)}/k_\omega$ with $|i_{max}| = 30$ A.
8. Speed controller output limit influences step response.
9. Step of load torque.
10. To show that the machine is able to work as a generator as well as a motor.

3 Comparison

Beside the documentation of the control design, the report for this unit should contain the following comparisons. Please keep in mind, that for the applied step changes the PI-controller should be able to bring the control error down to zero. So the interesting time interval for diagrams is normally very close around the step change.

3.1 Measured values

3.1.1 Current control

Compare q-current component of experiments

- 3 and 5
- 4 and 6

3.2 Simulation vs. measurement

3.2.1 Current control

The interesting quantities for comparison of measurement and simulation are current components (reference and actual values) and voltages (the current controllers output, compensation voltage and their limited sum). Please split up into separate diagrams for d- and q-voltage components.

Simulate the experiments

- 1, 2, 3 and 5

Speed control of the dc-machine is available and the whole drive system can be modelled (recommended). But it is also sufficient to apply a similar scheme as used for dc-machine tests.

To get beyond the abilities of the model structure used for controller design, this more detailed model has to include converter non-linearity, reference current limitation, compensation voltages, position quantisation due to incremental encoder and the rotor flux observer. Since converter non-linearity acts on phase values the transformations have to be included too.

The induction machine block in fig. 2 includes the electrical system only. Voltage equations might be implemented in stator reference frame. The only mechanical quantity is mechanical speed. This speed is available as filtered signal only and is a shortcoming of this model. The transformation angles ε and ε_{pre} are calculated from the increment of electrical rotorposition $\Delta\varphi_{el}$ and the estimated rotorflux position ρ . Please do not integrate filtered rotor speed to get rotor position, information will be lost by this way. Recorded signals (loaded into workspace) are discrete time quantities. If these signals are input to discrete time systems, linear interpolation has to be avoided. Uncheck "interpolate data" for inports or "from workspace" blocks.

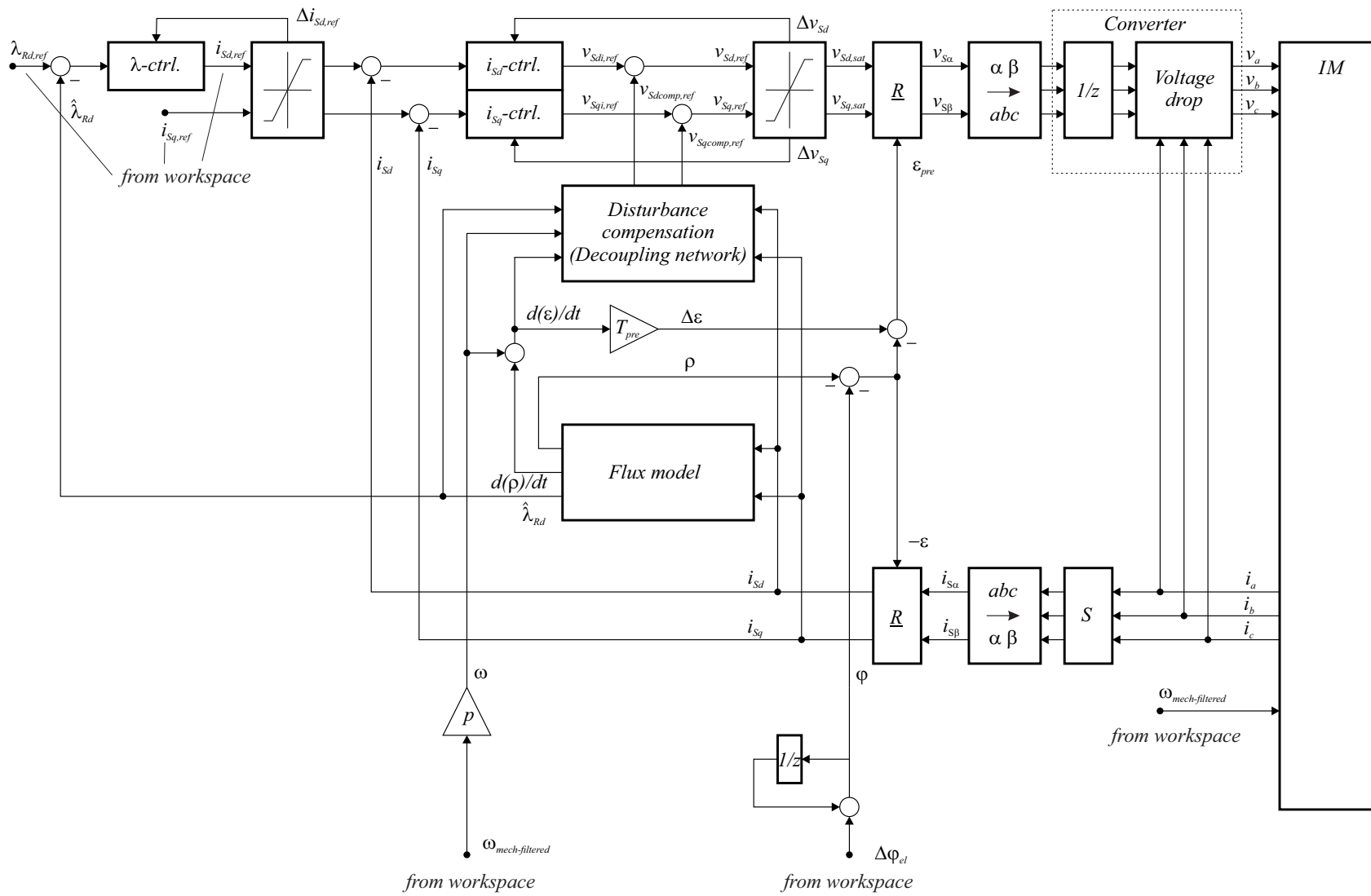


Figure 2: Model structure for current- and flux control simulation

3.2.2 Speed control

Now reference speed and actual speed have to be added to the interesting quantities from section 3.2.1.

Complete structure has to be modelled according to figure 3.

The input "load torque" can be calculated by multiplying the DC-machine armature current by torque constant $K\Phi$ of the DC-machine. Excitation current was set to 0.6 A.

Using the two body mechanical system, keep in mind that the incremental encoder is attached to the DC-machine shaft.

Simulate the experiments

- 7, 8, 9 and 10

For experiment 10 there is no armature current and no excitation current, DC-machine converter is used as a brake chopper.

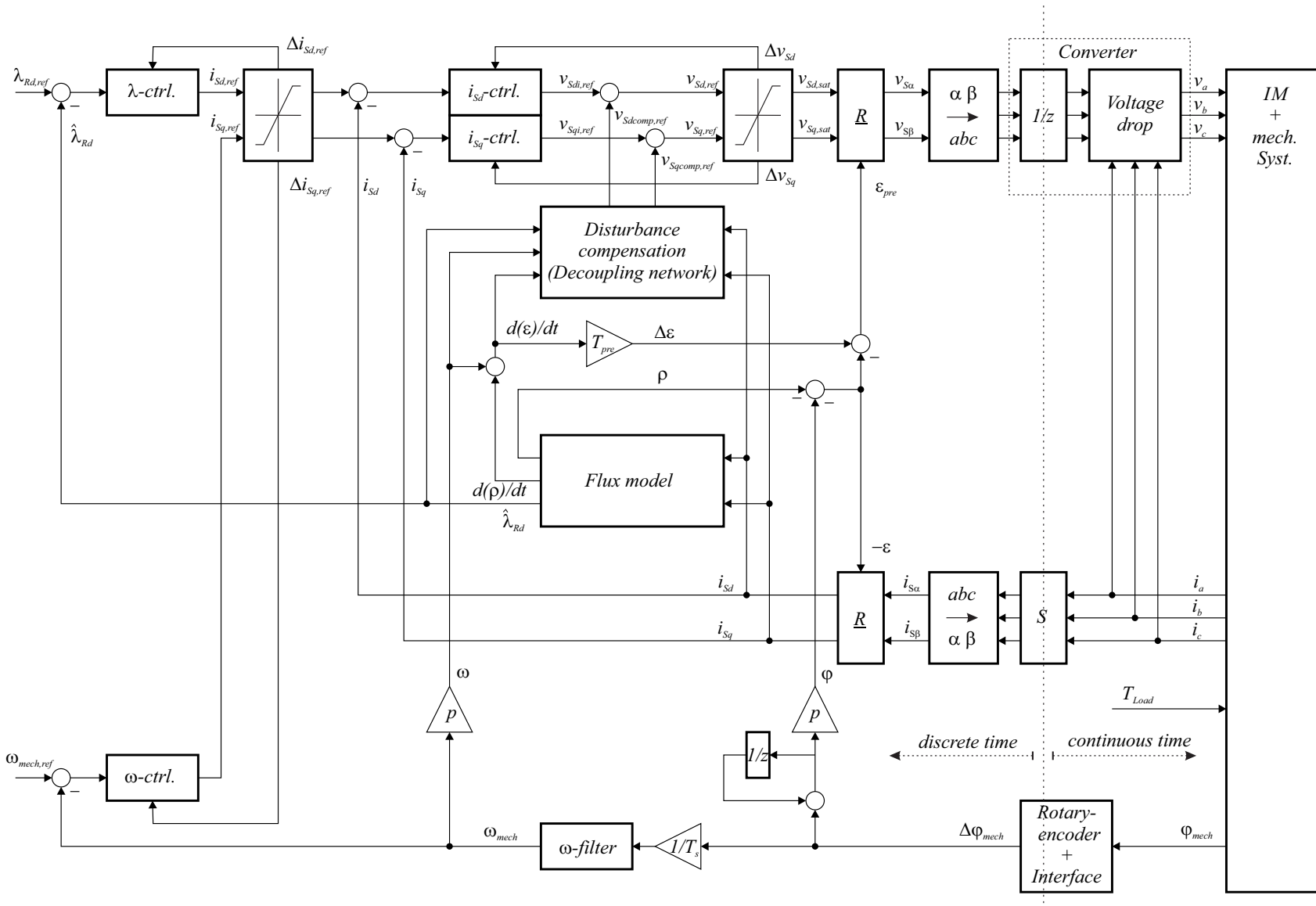


Figure 3: Model structure for speed control simulation

4 Signals

Name	Meaning
Model Root/mr/DCM/ia_controller/ref/Out1	$i_{a,ref}$ DC-machine
Model Root/mr/IM/Observer/al	ε
Model Root/mr/IM/Observer/alpha	ε_{pre}
Model Root/mr/IM/Observer/alphap	$\frac{d\varepsilon}{dt}$ stator frequency
Model Root/mr/IM/Observer/isd	i_{sd}
Model Root/mr/IM/Observer/isq	i_{sq}
Model Root/mr/IM/Observer/dphi/Out1	$\Delta\varphi_{el}$
Model Root/mr/IM/i-ctrl/Demux3/Out1	$v_{sd,comp,ref}$
Model Root/mr/IM/i-ctrl/Demux3/Out2	$v_{sq,comp,ref}$
Model Root/mr/IM/i-ctrl/id/u	$v_{sdi,ref}$
Model Root/mr/IM/i-ctrl/iq/u	$v_{sqi,ref}$
Model Root/mr/IM/i-ctrl/vsat/Switch2/Out1	$v_{sd,sat}$
Model Root/mr/IM/i-ctrl/vsat/Switch3/Out1	$v_{sq,sat}$
Model Root/mr/IM/psir-ctrl/act/Out1	$\widehat{\lambda}_{Rd}$
Model Root/mr/IM/psir-ctrl/ref/Out1	$\widehat{\lambda}_{Rd,ref}$
Model Root/mr/IM/ref_distribution/isd_soll1/Out1	$i_{sd,ref}$
Model Root/mr/IM/ref_distribution/isq_soll1/Out1	$i_{sq,ref}$
Model Root/mr/IM/w-ctrl/act/Out1	ω_{mech} (filtered)
Model Root/mr/IM/w-ctrl/ref/Out1	$\omega_{mech,ref}$
Model Root/mr/IO/scale/dcm_ia	i_a DC-machine
Model Root/mr/IO/scale/torque	shaft torque
Model Root/mr/IO/vdc_filter/Out1	filtered DC-link voltage

Table 2: Meaning of the recorded signals